Advanced Cu Surface Modification Using Ar-CH_4 Plasma Pre-treatment for Direct Bonding Applications

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ABSTRACT

AI applications utilizing large language models and transformer architectures have become essential technologies, where interconnection speed between logic and memory components has emerged as the primary performance bottleneck. High Bandwidth Memory (HBM) stacking technology enables parallel computation for AI applications but faces challenges in thermal management, electrical performance, and structural integrity at increased stack heights. Hybrid bonding technology addresses these issues through reduced interconnect length and superior thermal-mechanical properties compared to solder bumps, though challenges remain in Cu surface antioxidation, dishing control, and low-temperature bonding.

This study investigates reductive Ar-CH_4 plasma pre-treatment for direct Cu-to-Cu bonding using comprehensive characterization techniques including XPS, TOF-SIMS, TEM, and sheet resistance measurements. The treatment mechanism involves hydrogen ions and radicals reducing Cu oxides, followed by C_xH_y species deposition forming the protective layer that prevents re-oxidation in ambient air.

Design of experiments (DOE) optimization based on XPS analysis identified optimal conditions of high power, low pressure, and high CH_4 flow rate, effectively removing 1-2 nm native Cu oxide while depositing ~6 nm protective hydrogenated amorphous carbon (a-C:H) layer. Sheet resistance measurements revealed no electrical degradation following plasma treatment, with values equal to or improved compared to untreated surfaces, confirming that the a-C:H layer does not compromise electrical properties. Direct Cu-to-Cu bonding was performed at 260°C under 15 MPa for 1 hour, followed by annealing at 200°C for 1 hour. XPS and TEM analysis revealed that residual surface oxygen significantly influences bonding mechanisms: complete oxide removal resulted in carbon-to-carbon bonding with complete absence of oxygen at the interface and slight carbon diffusion into Cu subsurface, while samples with remaining oxides exhibited diffusion-based bonding mechanisms accompanied by void formation and carbon clustering. Shear test results demonstrated superior mechanical strength compared to non-plasma treated samples. These results indicate the feasibility of Cu-to-Cu bonding for future HBM applications.

Keywords: Cu-to-Cu direct bonding, reductive plasma treatment, low temperature bonding.

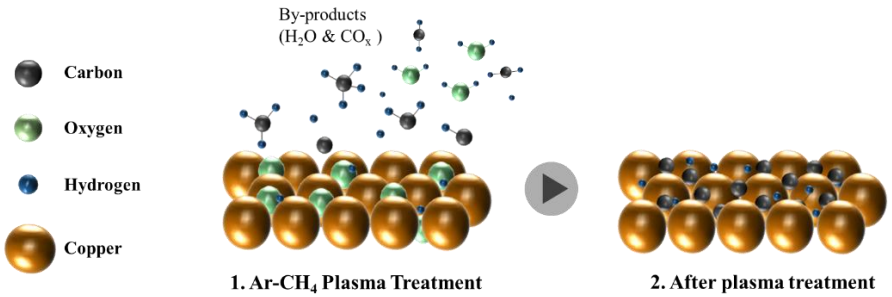


Fig. 1. The Schemetic Illustration of Ar-CH_4 Plasma Treatment